

KITAKYUSHU AP, Japan

WMO: 478530

Lat: 33.846N

Lon: 131.035E

Elev: 6

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.0	-8.0	1.9	3.0	-6.8	2.1	4.0	13.4	3.5	12.4	3.8	6.7	290	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	32.2	25.5	31.1	25.6	30.1	25.3	26.7	29.9	26.3	29.5	26.0	29.2	4.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	28.5	25.2	20.3	28.4	25.0	20.0	28.4	83.9	30.1	82.1	29.6	80.4	29.2	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.3	9.2	-0.9	34.4	1.7	1.0	-2.1	35.1	-3.1	35.6	-4.0	36.2	-5.2	36.9		
			-2.8	27.5	1.5	0.6	-3.9	27.9	-4.8	28.3	-5.6	28.7	-6.7	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	6.4	7.1	10.0	14.2	18.9	22.4	26.7	28.3	24.9	20.0	14.2	9.0
	DBStd	7.79	2.34	2.55	2.62	2.43	2.03	1.83	2.20	1.80	2.15	2.28	2.89	2.70
	HDD10.0	285	112	87	32	2	0	0	0	0	0	0	2	51
	HDD18.3	1506	368	316	258	124	18	0	0	0	0	9	125	289
	CDD10.0	2808	1	5	32	128	276	372	517	568	446	311	129	21
	CDD18.3	987	0	0	0	1	36	122	259	310	196	62	2	0
	CDH23.3	7706	0	0	1	3	53	337	2365	3531	1309	107	1	0
	CDH26.7	2253	0	0	0	0	4	34	688	1300	223	3	0	0
Wind	WSAvg	4.1	4.4	4.1	4.2	4.1	3.9	3.7	3.7	4.2	4.1	4.0	3.9	4.6
Precipitation	PrecAvg	1951	62	81	121	144	168	369	318	198	223	119	89	60
	PrecMax	3102	181	157	340	256	305	690	692	389	508	361	196	126
	PrecMin	1168	18	27	46	41	55	71	71	53	49	14	23	4
	PrecStd	382	37	33	56	53	64	154	178	103	131	88	46	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	15.9	19.9	22.8	27.0	29.1	33.2	34.1	31.1	27.0	21.8	16.9
		MCWB	9.7	12.9	14.4	14.3	17.9	23.2	25.4	25.7	24.8	21.8	17.5	14.2
	2%	DB	11.9	13.2	16.9	20.2	24.9	27.8	32.0	32.9	29.9	25.8	20.8	15.0
		MCWB	8.6	10.3	12.3	14.0	17.0	22.5	25.1	25.8	24.1	21.5	17.1	12.2
	5%	DB	10.9	12.0	15.1	19.0	23.1	26.1	30.9	31.8	28.9	24.2	19.8	14.0
		MCWB	7.8	9.2	11.5	13.9	17.5	21.7	25.4	25.8	24.0	20.0	16.3	11.1
	10%	DB	9.9	10.9	13.9	17.9	22.1	25.1	29.9	30.9	28.0	23.2	18.8	13.0
		MCWB	7.3	8.3	10.8	13.5	17.3	21.3	25.2	25.8	23.6	18.9	15.5	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	13.5	15.3	17.1	20.8	24.7	27.2	27.5	26.4	23.7	19.4	15.3
		MCDB	13.0	15.7	18.2	19.5	22.9	27.5	29.8	30.4	28.8	25.5	20.8	16.3
	2%	WB	9.7	11.3	13.3	16.0	20.0	23.7	26.6	26.8	25.6	22.4	18.3	13.0
		MCDB	10.6	12.9	15.4	18.4	21.9	25.8	29.4	30.2	28.2	24.8	19.8	14.6
	5%	WB	8.6	10.0	12.3	15.3	19.3	23.1	26.1	26.5	25.0	21.1	17.1	11.8
		MCDB	10.0	11.3	14.5	17.6	21.5	25.3	29.2	29.9	27.8	23.7	18.8	13.5
	10%	WB	7.5	8.8	11.3	14.7	18.5	22.4	25.7	26.2	24.3	19.8	15.7	10.8
		MCDB	9.4	10.4	13.5	16.9	21.0	24.4	28.8	29.6	27.2	22.6	18.0	12.7
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.9	5.6	5.7	5.7	4.4	4.3	4.5	4.4	4.8	4.8	4.5
		MCDBR	5.8	6.4	7.6	7.7	7.5	6.1	5.9	5.6	4.9	5.3	5.4	5.4
		MCWBR	4.2	4.8	5.0	4.3	4.0	3.1	2.4	2.4	2.6	3.1	3.9	4.5
	5% WB	MCDBR	5.2	5.6	6.8	5.8	5.6	4.4	4.7	4.6	4.1	4.4	4.4	4.6
		MCWBR	4.6	5.0	5.2	4.1	3.6	2.7	2.3	2.3	2.5	3.5	3.9	4.6
Clear-Sky Solar Irradiance	taub	0.428	0.469	0.531	0.553	0.558	0.571	0.570	0.534	0.502	0.457	0.430	0.420	
	taud	2.066	1.992	1.845	1.846	1.895	1.946	1.994	2.094	2.136	2.164	2.152	2.102	
	Ebn at Noon	748	761	746	754	755	742	740	761	765	766	743	729	
	Edh at Noon	131	157	197	205	198	188	178	158	146	131	120	120	
All-Sky Solar Radiation	RadAvg	2.17	2.88	3.96	4.81	5.24	4.50	4.88	5.12	4.07	3.43	2.55	2.00	
	RadStd	0.20	0.28	0.28	0.42	0.59	0.51	0.73	0.56	0.42	0.36	0.23	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page